

EDUCATION

University of Massachusetts Amherst 2021 – Current

M.S. & Ph.D. in Computer Science

Advisor: Dr. Donghyun Kim

Research Focus: Visual Navigation, Human-Robot Interaction, Assistive Robotics

Hanyang University ERICA 2013 – 2019

B.S. in Robot Engineering (Cum Laude)

Exchange Student: The University of Texas at Austin, Dept. of ECE 2017 – 2018

HONORS AND AWARDS

- **Best Paper Finalist**, IEEE International Conference on Robotics and Automation (ICRA) 2026
- **Honourable Mention**, ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026
- **Best Paper Finalist**, International Conference on Ubiquitous Robots (UR) 2024
- **Best Paper Award**, ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) 2024
- **Robert and Deanna Hagerty Robotics Scholarship**, UMass Amherst 2024
- **Vision Assistance Race 2nd place**, CYBATHLON Challenges 2023
- **UMass Amherst CICS Jumpstart Fellowship**, UMass Amherst 2021

PUBLICATIONS

Conference

1. **GuideTWSI: A Diverse Tactile Walking Surface Indicator Dataset from Synthetic and Real-World Images for Blind and Low-Vision Navigation.** H. Hwang, S. Yang, A. N.H. Nguyen, P. Goel, K. Adhikari, S.I. Lee, J. Biswas, N.A. Giudice, D. Kim. **ICRA 2026. [Best Paper Finalist]**
2. **Human-Centered Development of Guide Dog Robots: Quiet and Stable Locomotion Control.** S. Yu[†], H. Hwang[†], T.M. Dang, J. Biswas, N.A. Giudice, S.I. Lee, D. Kim. **ICRA 2026. [Oral]**
3. **GuideNav: User-Informed Development of a Vision-Only Robotic Navigation Assistant For Blind Travelers.** H. Hwang, S. Yang, J.S. Monon, N.A. Giudice, S.I. Lee, J. Biswas, D. Kim. **HRI 2026. [Honourable Mention, 3.6%]**
4. **Synthetic data augmentation for robotic mobility aids to support blind and low vision people.** H. Hwang, K. Adhikari, S. Shodhaka, D. Kim. **RiTA 2024.**
5. **Is it safe to cross? Interpretable Risk Assessment with GPT-4V for Safety-Aware Street Crossing.** H. Hwang, S. Kwon, Y. Kim, D. Kim. **UR 2024. [Best Paper Finalist, Top 8]**
6. **Towards Robotic Companions: Understanding Handler-Guide Dog Interactions for Informed Guide Dog Robot Design.** H. Hwang, H.T. Jung, N.A. Giudice, J. Biswas, S.I. Lee, D. Kim. **CHI 2024. [Best Paper Award, < 1%]**
7. **System Configuration and Navigation of a Guide Dog Robot: Toward Animal Guide Dog-Level Guiding Work.** H. Hwang[†], T. Xia[†], I. Keita, K. Suzuki, J. Biswas, S.I. Lee, D. Kim. **ICRA 2023.**
8. **Control Scheme and Uncertainty Considerations for Dynamic Balancing of Passive-Ankled Bipeds and Full Humanoids.** D. Kim, S.J. Jorgensen, H. Hwang, L. Sentis. **Humanoids 2018.**
9. **Computationally-Robust and Efficient Prioritized Whole-Body Controller with Contact Constraints.** D. Kim, J. Lee, O. Campbell, H. Hwang, L. Sentis. **IROS 2018.**

Journal

1. **ElderSim: A Synthetic Data Generation Platform for Human Action Recognition in Eldercare Applications.** H. Hwang, C. Jang, G. Park, J. Cho, I.J. Kim. **IEEE Access** **2021**.
2. **Sensitive Capacitive Pressure Sensors over a Wide Pressure Range Enabled by the Hybrid Responses of a Highly Porous Nanocomposite.** K.H. Ha, W. Zhang, H. Jang, S. Kang, L. Wang, P. Tan, H. Hwang, N. Lu. **Advanced Materials** **2021**.

RESEARCH EXPERIENCE

University of Massachusetts Amherst, Dynamic & Autonomous Robotics Lab 2021 – Current
Graduate Research Assistant

- User-centered guide dog robot development for blind and low-vision individuals: Foundation model-based perception and VLA-based planning in legged systems for safe and efficient navigation.
- VLM evaluation for safe street crossing and synthetic data augmentation for perception model training.

The University of Texas at Austin, Human Centered Robotics Lab 2017 – 2018
Undergraduate Research Assistant

- Testing and optimizing the 6 DOF passive-ankled biped robot, Mercury, for walking.

WORK EXPERIENCE

Amazon – Frontier AI & Robotics, San Francisco, USA 2026 – Current
Member of Technical Staff Intern

- Developing VLA models for navigation to effectively leverage implicit map representations.

Glidence, Seattle, USA 2025 – 2026
PhD Research Intern

- Developed robust vision-based visual teach-and-repeat and VLM-based guidance for robot navigation.

Korea Institute of Science and Technology – Center for AI, Republic of Korea 2019 – 2020
Research Intern

- Real-time human action recognition system development and synthetic data augmentation evaluation.

MENTORSHIP & TEACHING

Mentorship, UMass Amherst 2023 – Present
Advised 20+ students across various research programs:

- **Early Research Scholars Program:** Mentor for 12 undergraduate researchers on perception projects.
- **Thesis & Independent Study:** Advised 10+ students; notably mentored one student to **1st Place** at ACM TAPIA '24 and another to a National Conference presentation.

Teaching Assistant, UMass Amherst 2022 – 2024
Robotics (CS603), Intro. to Robotics (CS403), Reasoning Under Uncertainty (CS240)

PROFESSIONAL SERVICE & OUTREACH

Student Representative: IEEE Robotics & Automation Society TC on HRI & Coordination 2026

Peer Review: *Robotics:* ICRA, IROS, RA-L, UR, RiTA, Humanoids. *HRI:* THRI, HRI, CHI, ACL.

Outreach & Demonstrations: 2022 – Present

Led autonomous guide robot demos for the **Carroll Center for the Blind**, **Fidelco Guide Dog Foundation**, **Foundation Fighting Blindness**, **Polus Center**, and the **Massenberg STEM Institute**.

Leadership: President, UMass Korean Graduate Student Association (KGSA) 2022 – 2023